K241033 CN LAB#8 tasks

Step 1&2 : I have linux installed already in my laptop with all the tools etc so this step is unneeded

Step 2:

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8$ git clone https://gitlab.com/nsnam/ns-3-dev.git
cd ns-3-dev
Cloning into 'ns-3-dev'...
remote: Enumerating objects: 233452, done.
remote: Counting objects: 100% (5166/5166), done.
remote: Compressing objects: 100% (1854/1854), done.
Receiving objects:  14% (32684/233452), 15.77 MiB | 1.47 MiB/s
```

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ ./ns3 configure --enable-examples
Warn about uninitialized values.
-- The CXX compiler identification is GNU 16.2.1
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Using default output directory /home/raed/Documents/dev/assignments/cn_lab_8/ns-3-dev/build
-- Performing Test GCC_WORKING_PEDANTIC_SEMICOLON
-- Performing Test GCC_WORKING_PEDANTIC_SEMICOLON - Success
```

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ ./ns3 build\
>
[0/2] Re-checking globbed directories...
[13/1628] Building CXX object src/core/CMakeFiles/core.dir/model/synchronizer.cc.o
```

Step 3:

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ ./ns3 run hello-simulator
[0/2] Re-checking globbed directories...
ninja: no work to do.
Hello Simulator
```

Step 4:

```
Hello Simulator
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ touch scratch/first.cc
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ nano scratch/first.cc
```

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ ./ns3 run scratch/first.cc
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/raed/Documents/dev/assignments/cn_lab_8/ns-3-dev/build/scratch/ns3-dev-first-default
At time +2s client sent 1024 bytes to 10.1.1.2 port 9
At time +2.00369s server received 1024 bytes from 10.1.1.1 port 49153
At time +2.00369s server sent 1024 bytes to 10.1.1.1 port 49153
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
```

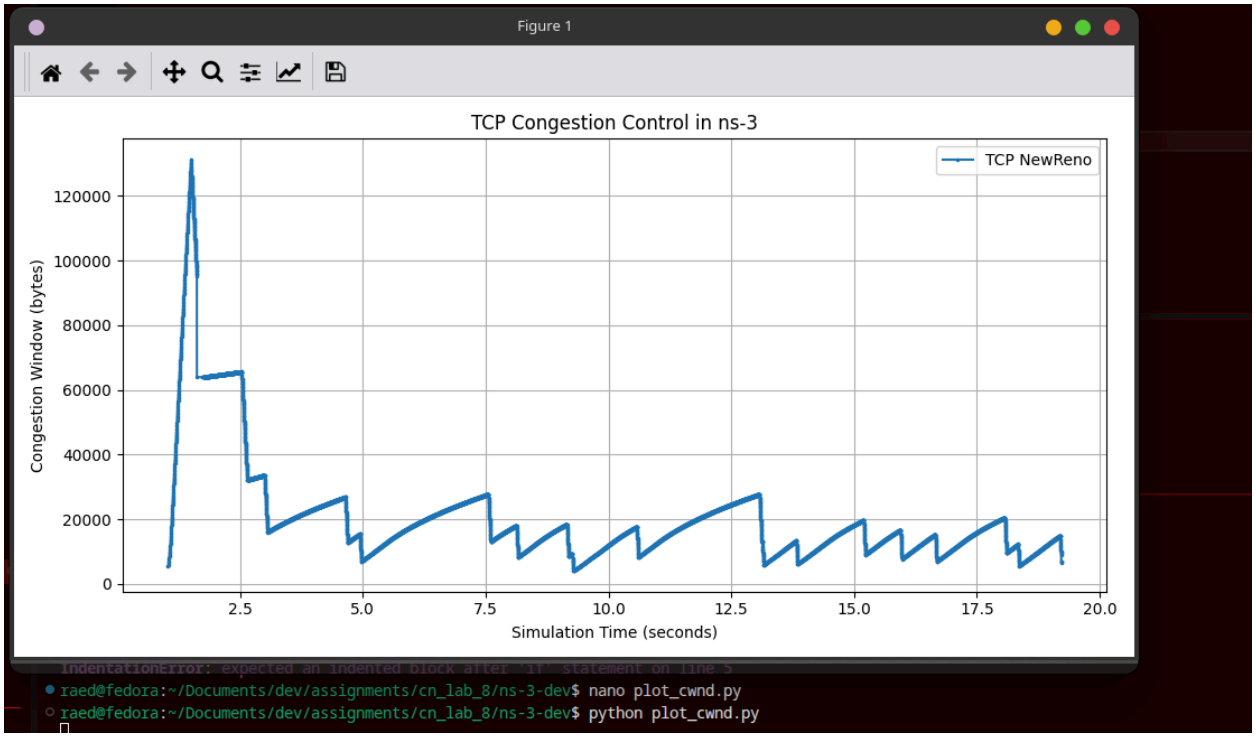
Step 5:

```
At time +2.00737s client received 1024 bytes from 10.1.1.2 port 9
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ touch scratch/tcp-congestion.cc
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ nano scratch/tcp-congestion.cc
```

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ ./ns3 run scratch/tcp-congestion.cc
[0/2] Re-checking globbed directories...
[2/3] Linking CXX executable /home/raed/Documents/dev/assignments/cn_lab_8/ns-3-dev/build/scratch/ns3-dev-tcp-congestion-default
```

```
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ head cwnd.dat
1.02019 5360
1.0413  5896
1.04319 6432
1.04508 6968
1.04697 7504
1.04885 8040
1.06233 8576
1.06422 9112
1.06611 9648
1.068   10184
raed@fedora:~/Documents/dev/assignments/cn_lab_8/ns-3-dev$ wc -l cwnd.dat
9137 cwnd.dat
```

step 6:



Step 7:

The graph initially showed a slow increase in the congestion window then at the first peak congestion was detected (3ack of same sequence) and to avaoid it cwnd was reduced then immediately after a few iterations congestion was detected again at the second peak and then the cwnd was reduced

again. The saw tooth pattern can be detected here since the congestion window slowly increases then congestion is detected then congestion window falls down and this happens in a loop. Every peak represents the detection of a congestion(3ack for same sequence)